# SQUID: A Bayesian Approach for Physics-Informed Event Modeling

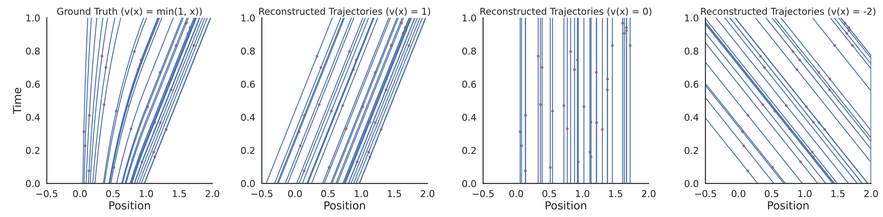
Sumantrak Mukherjee, Sebastian Vollmer, Gerrit Grossmann
sumantrak.mukherjee@dfki.de

github.com/gerritgr/squid

## Motivation

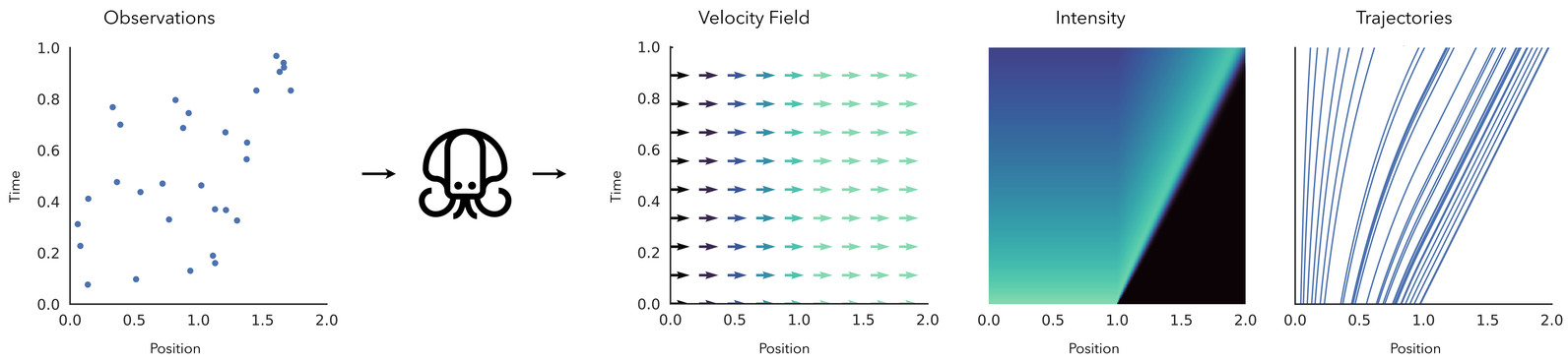
- Spatio-temporal events (e.g., epidemiology, crime, ecology, climate) exhibit intensity drift over time.
- Classical STPP models estimate intensity directly, often lacking interpretability and uncertainty.
- Need a method to infer underlying mechanisms driving spatial movement of intensities.

## Problem Setting



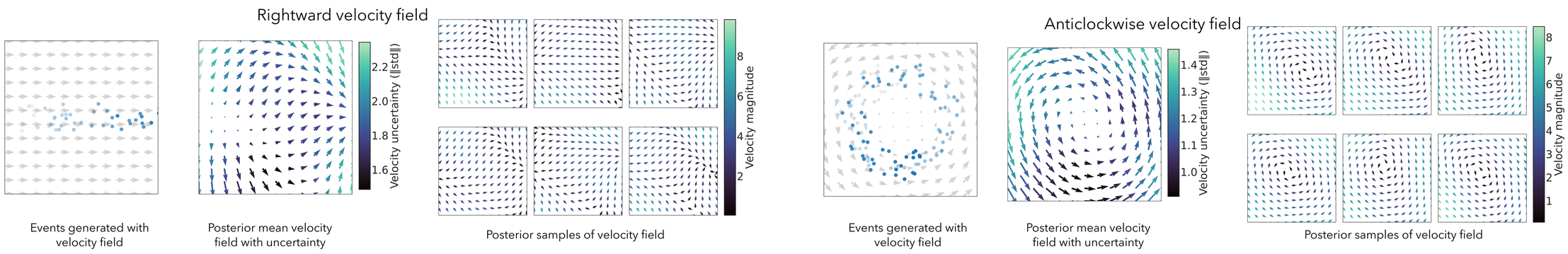

- Input: List of events (x,y,t) in space and time
- Output: Vector field to represent drift
- Specialty: Use posterior distribution over the vector field
- Framework: Vector field transports the intensity in a spatio-temporal point process
- Challenge: Likelihood evaluation requires costly PDE solves

## Method




- Sample a velocity field from the GP.
- Trick 1: Move all events along the velocity field following their ODE (instead of solving the PDE).
- Trick 2: Use local trajectory density as a surrogate likelihood.
- Update the posterior using a fully differentiable inference pipeline, enabling gradient-based sampling or optimization.

## Results




## Future Work

- Extend to time varying velocity fields.
- Apply to real world data.
- Incorporate contextual data for vector field prediction.
- Disentangle local self-excitation effects from global drift patterns.

## Conclusion

SQUID provides an efficient, interpretable, and uncertainty-aware Bayesian framework for recovering the latent velocity fields that explain the spatio-temporal intensity drift.

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